

Logistic Regression Analysis: Philippine Mouse Deer Detectability

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```
# 1. Load libraries and read data
```

```
library(terra)
```

```
## terra 1.9.34
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.3      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x tidyr::extract() masks terra::extract()
```

```
## x dplyr::filter()  masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
setwd("C:/Users/Admin/downloads")
```

```
species_data <- read.csv("ph_mouse_deer.csv")
```

```
species_clean <- species_data[complete.cases(species_data$Detected), ]
```

```
species_sf <- st_as_sf(species_clean, coords = c("Longitude", "Latitude"), crs = 4326)
```

```
species_ext <- ext(species_sf)
```

```
buffer <- 0.5
```

```
zoom_extent <- ext(
```

```
  xmin(species_ext) - buffer, xmax(species_ext) + buffer,
```

```
  ymin(species_ext) - buffer, ymax(species_ext) + buffer
```

```
)
```

```
if(file.exists("PHL_msk_alt.tif")) {
```

```

# Load the raster if it hasn't been loaded yet
map_raster <- rast("PHL_msk_alt.tif")
print("Raster successfully loaded.")
} else {
  stop("Error: 'PHL_msk_alt.tif' not found in your working directory.")
}

```

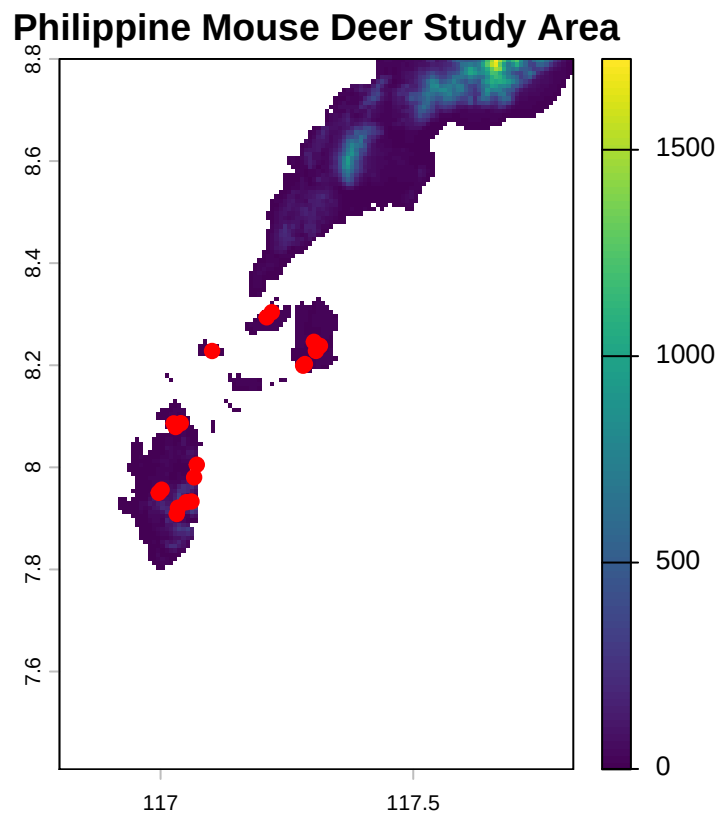
```
## [1] "Raster successfully loaded."
```

```

map_crop <- crop(map_raster, zoom_extent)

plot(map_crop, main = "Philippine Mouse Deer Study Area")
plot(st_geometry(species_sf), add = TRUE, col = "red", pch = 19)

```



```

species_clean$HabitatCode <- as.factor(species_clean$HabitatCode)
model_log <- glm(Detected ~ AverageDistance + Effort + HabitatCode,
  data = species_clean,
  family = binomial)
summary(model_log)

```

```

##
## Call:
## glm(formula = Detected ~ AverageDistance + Effort + HabitatCode,
##      family = binomial, data = species_clean)

```

```
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -19.57441  3525.85824  -0.006   0.996
## AverageDistance    0.16806    0.12985   1.294   0.196
## Effort          0.05477    1.10880   0.049   0.961
## HabitatCode2     19.79372  3525.85719   0.006   0.996
## HabitatCode3     16.51916  3525.85736   0.005   0.996
## HabitatCode4     36.08148  7414.61276   0.005   0.996
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 26.287  on 18  degrees of freedom
## Residual deviance: 14.638  on 13  degrees of freedom
## AIC: 26.638
##
## Number of Fisher Scoring iterations: 17
```

```
par(mfrow=c(1,2))

dist_range <- seq(min(species_clean$AverageDistance),
                  max(species_clean$AverageDistance), length.out = 100)

pred_data_dist <- data.frame(AverageDistance = dist_range,
                             Effort = mean(species_clean$Effort),
                             HabitatCode = factor(rep(levels(species_clean$HabitatCode)[1], 100)))

probs_dist <- predict(model_log, newdata = pred_data_dist, type = "response")

plot(species_clean$AverageDistance, species_clean$Detected,
     col = "red", pch = 19, xlab = "Average Distance", ylab = "Probability",
     main = "Detection Probability vs Distance")
lines(dist_range, probs_dist, col = "blue", lwd = 2)

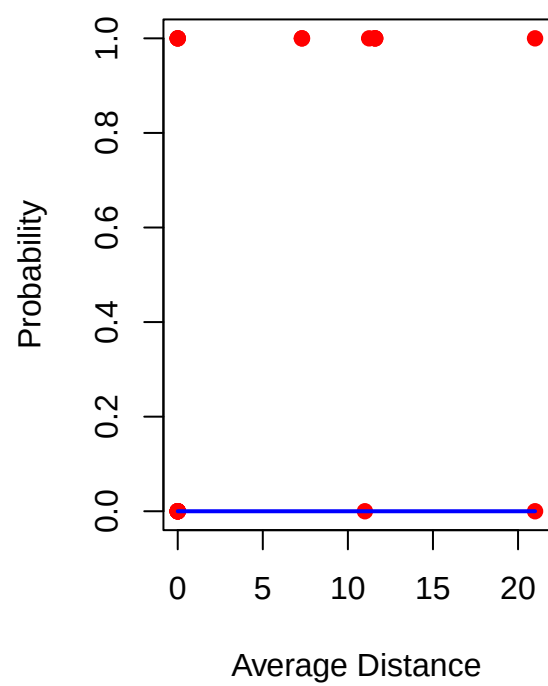
effort_range <- seq(min(species_clean$Effort),
                   max(species_clean$Effort), length.out = 100)

pred_data_eff <- data.frame(AverageDistance = mean(species_clean$AverageDistance),
                             Effort = effort_range,
                             HabitatCode = factor(rep(levels(species_clean$HabitatCode)[1], 100)))

probs_eff <- predict(model_log, newdata = pred_data_eff, type = "response")

plot(species_clean$Effort, species_clean$Detected,
     col = "red", pch = 19, xlab = "Sampling Effort", ylab = "Probability",
     main = "Detection Probability vs Effort")
lines(effort_range, probs_eff, col = "green", lwd = 2)
```

Detection Probability vs Distance



Detection Probability vs Effort

